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Improved estimation for just-noticeable visual distortion

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Abstract

Perceptual visibility threshold estimation, based upon characteristics of the human visual system (HVS), has wide applications in digital image/video processing. An improved scheme for estimating just-noticeable distortion (JND) is proposed in this paper. It is proved to outperform the DCTune model, with the major contributions of a new formula for luminance adaptation adjustment and the incorporation of block classification for contrast masking. The HVS visibility threshold for digital images exhibits an approximately parabolic curve versus gray levels and this has been formulated to yield a more accurate base threshold. Moreover, edge regions have been differentiated via block classification to effectively avoid over-estimation of JND in the said regions. Experiments with different images and the associated subjective tests show improved performance of the proposed scheme over the DCTune model for luminance adaptation (especially in dark regions) and masking effect in edge regions. Our model has further demonstrated to achieve favorable results in perceptual visual distortion gauge and image compression. The improvement in JND estimation facilitates better visual distortion measurement and visual signal compression.

Keywords: HVS; JND; DCT; Visibility threshold; Image compression; Visual distortion metric

1. Introduction

Digital images are acquired, enhanced, compressed and reconstructed for various applications, and more often than not, human eyes are the ultimate receiver of the processed images. The human visual system (HVS) is only capable of perceiving the pixel change above a certain visibility threshold determined by the underlying physiological and psychophysical mechanisms. The JND [12] refers to the minimum visibility threshold when visual contents can be distinguished. The JND can be decided with subjective viewing tests. For instance, one JND can be defined as the level at which statistically 75% of the viewers can sense the difference [27]. However,

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subjective tests are time consuming and not suitable for most practical image processing applications.

Methods of automatic JND threshold[1] derivation have long been research topics of images and video [1–4,9–12,22,25–27,30,33–35], mainly based upon temporal/spatial contrast sensitivity function (CSF), background luminance adaptation and contrast masking.[2] A JND map of the image can facilitate adaptive image compression [1–3,9–12,22,23,25,30,33,34], perceptual visual quality evaluation [4,20,26] and adaptive watermarking [35].

The JND can be determined for either pixels or subbands. In [2,3], pixel-based JND thresholds are derived from background luminance adaptation and local spatial contrast masking. Subband-based JND [1,9–11,22,25,30,33,34] is of particular interest since the CSF can be more easily incorporated and most images are subband coded.

One of the early methods to decide subband-based JND was proposed in [25] by adjusting base sensitivity for each subband due to brightness adaptation and texture masking based upon averaged AC energy in each of the 16 subbands decomposed by the generalized quadrature mirror filter (GQMF). In [26], a JND map is generated with pyramid image decomposition and spatial filtering. In [9], the inter-channel masking effect of different orientations and frequencies was considered. In Daly's visible differences predictor (VDP) [4], the CSF acts as a frequency normalization process, and a modified Cortex transform [32] is then used in modeling the radial frequency selectivity and the orientational selectivity; after that a masking function is incorporated in each cortex band. In [34], the JND threshold was exploited with spatial frequency and orientation for wavelet subbands.

More intensive research in subband JND has been performed in DCT domain [1,11,22,30,33], because of DCT's popularity in image and video compression (e.g., JPEG, H.261/3/4, MPEG 1/2/4, and the emerging H.264 schemes). DCT-based JND is usually formulated as the product of a base threshold and an elevation factor. The former is usually determined by the spatial CSF and luminance adaptation, while the latter accounts for contrast masking.

A well-cited scheme for the perceptual threshold was developed by Ahumada and Peterson [1] with DCT decomposition, based on a spatial CSF that reflects the HVS' sensitivity versus spatial frequencies. The scheme has become the basis for a number of improved or simplified models [11,30,33], and has been extended for color images [22]. The basic scheme was improved into the DCTune model [33] by Watson after luminance adaptation effect had been added to the base threshold and contrast masking [15,14] had been calculated as the elevation factor. Luminance adaptation accounts for the masking from local background luminance, while contrast masking denotes the visibility reduction of one visual feature at the presence of another one. More recently, the locally-adaptive perceptual image coder (LAPIC) has been developed by Hontsch and Karam in [11] based upon the DCTune model with a foveal region being considered instead of a single pixel. In [30], Cortex transform is used to map DCT coefficients for mimicking the HVS' contrast masking characteristics more closely.

In the Ahumada–Peterson model [1], DCTune model [33] and its variation [11], the base threshold is not modeled accurately for luminance adaptation in dark and bright regions of a digital image because the local image properties [17] have not been fully exploited, and the elevation factor is not modeled for the contrast masking at or around edge regions. In the basic Ahumada–Peterson formula, linear luminance adaptation (i.e., Weber–Fechner's law) was implied [1,23,33], while in the simplified model [33], the power law for luminance adaptation was assumed. Moreover, the elevation factor [11,33] in the model causes the overestimation of the contrast masking effect at edge areas, because the power law for contrast discrimination [15,14] is oversimple to account for masking in real-world images. The HVS has lower visibility threshold at and around edges [6,7]. Tong and Venetsanopoulos classified all image blocks into edge, plain and texture regions [29] to differentiate the contrast masking. However, they

[1]Referred as *perceptual threshold* in some of the papers.

[2]Sometimes referred as *texture* or *activity* masking.

did not incorporate the spatial CSF and used merely a piecewise linear function for luminance adaptation.

This paper firstly proposes an improved formula for the base threshold in the DCTune model, by incorporating the HVS' sensitivity characteristics in dark and bright gray level regions of images. Secondly, the elevation factor is determined with classification of image blocks to avoid the over-estimation of the contrast masking effect around edges. The organization for the rest of this paper is as follows: Section 2 presents and illustrates the new DCT-based JND model for both the base threshold and the elevation factor; the use of a JND estimator is discussed in Section 3 for image distortion evaluation and image compression; in Section 4, our JND model is validated and compared with the DCTune model via specially-designed subjective viewing tests, and its performance in distortion gauge and image coding is also given; Section 5 concludes this paper.

2. Improved JND model

The JND in DCT domain is typically expressed as the product of a base threshold and an elevation factor [11,30,33]. Let n_1 and n_2 be the position indices of an $N \times N$ DCT block in an image, and i,j be the DCT subband indexes ($i,j = 0, 1, \ldots, N-1$). The corresponding JND can be expressed as

$$t_{\mathrm{JND}}(n_1, n_2, i, j) = t_{\mathrm{b}}(n_1, n_2, i, j) \times a_{\mathrm{e}}(n_1, n_2, i, j), \quad (1)$$

where the base threshold, $t_{\mathrm{b}}(n_1, n_2, i, j)$, accounts for spatial contrast sensitivity function (CSF) and background luminance adaptation, and the elevation factor, $a_{\mathrm{e}}(n_1, n_2, i, j)$, accounts for contrast masking in the image neighborhood. Note that N is usually 8.

2.1. Spatial CSF effect

The spatial CSF describes the effect of spatial frequency towards the HVS' sensitivity [18]. The absolute visibility threshold T is a function of background luminance L and spatial frequency f. There exists a minimum threshold $T_{\min}$ at spatial frequency f_p. In Ahumada–Peterson model [1,22], T for the (i,j)th DCT subband is determined as

$$T(i,j) = \frac{G}{\phi_i \phi_j (L_{\max} - L_{\min})} T^0(i,j), \quad (2)$$

where $L_{\max}$ and $L_{\min}$ are the display luminances corresponding to maximum and minimum gray levels (i.e., 255 and 0); G is the total number of gray levels ($G = 256$ for 8-bit image representation);

$$\phi_x = \begin{cases} \sqrt{\frac{1}{N}} & u = 0 \\ \sqrt{\frac{2}{N}} & u \neq 0 \end{cases}$$

being theDCT normalizing factor;

and

$$\log T^0(i,j) = \log \frac{s T_{\min}}{r + (1-r)\cos^2 \theta_{i,j}} + K(\log f_{i,j} - \log f_p)^2 \quad (3)$$

with

$$T_{\min} = \begin{cases} 0.142 \times \left(\frac{L}{13.45}\right)^{0.649}, & L \leqslant 13.45\ \mathrm{cd/m^2} \\ \frac{L}{94.7} & \text{otherwise} \end{cases} \quad (4)$$

$$f_p = \begin{cases} 6.78 \times \left(\frac{L}{300}\right)^{0.182}, & L \leqslant 300\ \mathrm{cd/m^2} \\ 6.78 & \text{otherwise} \end{cases} \quad (5)$$

$$K = \begin{cases} 3.125 \times \left(\frac{L}{300}\right)^{0.0706}, & L \leqslant 300\ \mathrm{cd/m^2} \\ 3.125 & \text{otherwise} \end{cases} \quad (6)$$

$$f_{i,j} = \frac{1}{2N_{\mathrm{DCT}}} \sqrt{\frac{i^2}{\omega_x^2} + \frac{j^2}{\omega_y^2}}, \quad (7)$$

$$\theta_{i,j} = \arcsin \frac{2 f_{i,0} f_{0,j}}{f_{i,j}^2}, \quad (8)$$

where ω_x and ω_y are the horizontal and vertical visual angles of a pixel, and they can be calculated based on viewing distance λ and the display width of a pixel Λ on the monitor:

$$\omega_\eta = 2\arctan\left(\frac{\Lambda_\eta}{2\lambda}\right), \tag{9}$$

where $\eta = x, y$.

Note that $T^0(0,0)$ takes the value of $\min\{T^0(1,0), T^0(0,1)\}$ to account for the fact that the decrease in sensitivity is slow at low spatial frequencies [1,13,23]. In Eq. (3), s and r are set to 0.25 and 0.6 to account for the *spatial summation* effect [22] and the *oblique* effect [1,22].

2.2. Adjustment for luminance adaptation

In Ahumada–Peterson and DCTune formulas [1,11,22,33], Weber–Fechner's law or its power law version was implied for luminance adaptation. Weber–Fechner's law states that the just-noticeable luminance contrast (T/L) remains constant over a luminance region [17,28]. However, this over-simplifies the viewing conditions for practical images. An observer is not only adapted to the background luminance, but also to the local image properties. The experimental result that is closer to typical image viewing conditions is shown in Fig. 1 [17], in which at each background luminance L_0 the contrast threshold exhibits a parabolic curve against local luminance L.

For digital images, much stronger dependence on local luminance L is exhibited than on background luminance L_0. This is mainly due to the following reason: digital images have been converted to a limited range of gray levels (typically just 256 levels) so the mean background luminance is not very different from one image to another; consequently, a single parabola curve in Fig. 1 can approximate the luminance adaptation in digital images. Other factors affecting luminance adaptation in digital images include the ambient illumination falling on the display and the γ-*correction* of the display tube that partially compensates for Weber–Fechner law's effect [19]. As a result, a higher visibility threshold T occurs in either very dark or very bright regions in an image, and a lower T occurs in regions with medium brightness. This is illustrated in Fig. 2 and is consistent with the findings of subjective viewing [3,10,12,19,25].

Since the local gray level luminance can be represented by the DC component of DCT for the block, $C(n_1, n_2, 0, 0)$, the effect of luminance adaptation can be determined as follows, in order to approximate the HVS characteristics shown

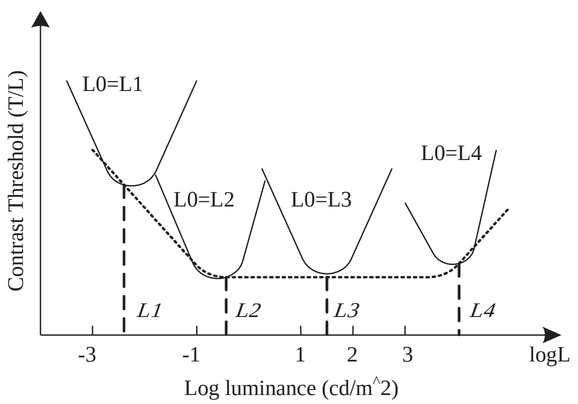

Fig. 1. Contrast threshold varies with local luminance L (L_0 is the background luminance, and T is the absolute threshold).

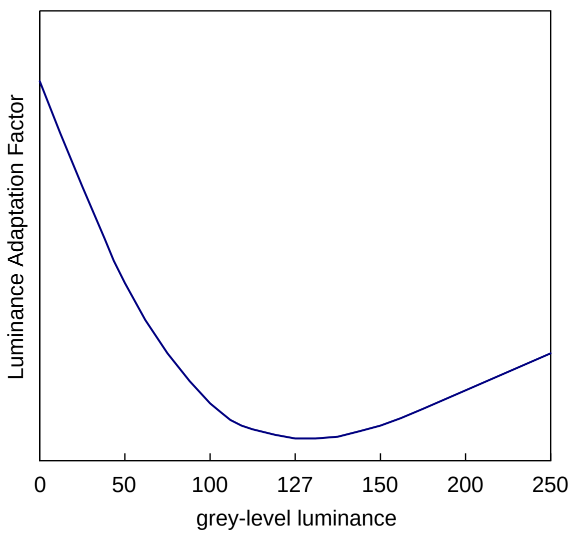

Fig. 2. Luminance adaptation for digital images.

in Fig. 2:

$$a_{\mathrm{lum}}(n_1,n_2) = \begin{cases} k_1\left(1-\frac{2C(n_1,n_2,0,0)}{GN}\right)^{\lambda_1}+1 & \text{if } C(n_1,n_2,0,0) \leqslant \frac{GN}{2} \\ k_2\left(\frac{2C(n_1,n_2,0,0)}{GN}-1\right)^{\lambda_2}+1 & \text{otherwise,} \end{cases} \tag{10}$$

where $k_1 = 2$, $k_2 = 0.8$, $\lambda_1 = 3$ and $\lambda_2 = 2$.

Fig. 3 compares our luminance adaptation scheme and the DCTune scheme [33] based upon Weber–Fechner's Law. As can be seen, our scheme adapts well to the HVS characteristics in low and high luminance regions in digital images; the major difference between the two models can be observed in the lower gray level (i.e., below 128) range.

With the compound effect of spatial CSF and luminance adaptation, the base threshold can be expressed as

$$t_{\mathrm{b}}(n_1,n_2,i,j) = T(i,j)a_{\mathrm{Lum}}(n_1,n_2). \tag{11}$$

2.3. Block classification for contrast masking

Contrast masking is an important phenomenon in the HVS perception [15,14] and refers to the reduction in the visibility of one visual component in the presence of another one. Usually noise becomes less visible in the presence of background patterns, especially in the regions where texture energy is high. On the contrary, noise is easily observed in smooth areas.

Texture energy is also high at object boundaries. However, the HVS has acute sensitivity at or near luminance edge in an image [7], probably because edge structure is simpler than a texture one and a typical observer has better prior knowledge on what edge should look like [6].

For more accurate evaluation of contrast masking, each DCT block is assigned to one of the three classes with descending order of the HVS sensitivity, namely, PLAIN, EDGE and TEXTURE [29]. Let L, M and H represent the sums of the absolute DCT coefficient values in the low-frequency (LF), medium-frequency (MF) and high-frequency (HF) groups, respectively, as shown in Fig. 4. The texture energy for the block is approximated by:

$$\mathrm{Tex}E = M + H. \tag{12}$$

Since the LF and MF portions of DCT coefficients reflect edge information, L/M and $(L+M)/H$ indicate the presence of edge [21,29]. We define:

$$E_1 = (\bar{L} + \bar{M})/\bar{H}, \tag{13}$$

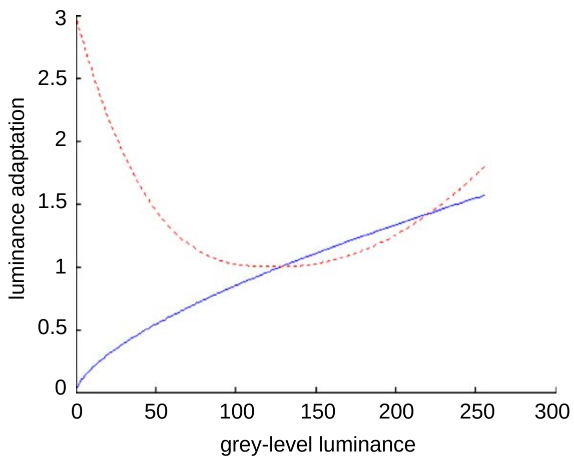

Fig. 3. Comparison of luminance adaptation schemes (solid line: Watson's luminance adaptation scheme; dashed line: our luminance adaptation scheme).

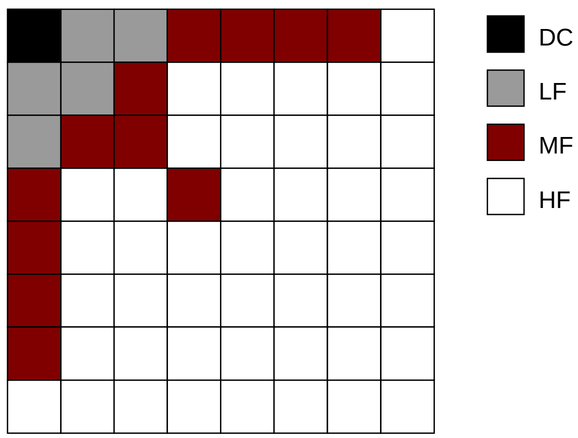

Fig. 4. DCT block classification.

and

$$E_2 = \bar{L}/\bar{M}, \tag{14}$$

where $\bar{L}$, $\bar{M}$ and $\bar{H}$ denote the means of L, M and H, respectively.

A block is classified as an EDGE one if

$$E_1 \geqslant \upsilon \tag{15a}$$

or

$$\max\{E_1, E_2\} \geqslant \alpha \quad \& \quad \min\{E_1, E_2\} \geqslant \beta, \tag{15b}$$

where $\alpha > \beta$.

Based upon Tex$E$, there are four cases as follows in block classification.

Case 1: Tex$E \leqslant \mu_1$. Since the texture energy is low, the block is a PLAIN class.

Case 2: $\mu_1 <$ Tex$E \leqslant \mu_2$. If (15) is met for $\alpha = \alpha_1$ and $\beta = \beta_1$, the block is assigned to EDGE class; otherwise it is assigned to PLAIN class.

Case 3: $\mu_2 <$ Tex$E \leqslant \mu_3$. If (15) is met for $\alpha = \alpha_1$ and $\beta = \beta_1$, the block is assigned to EDGE class; otherwise it will be assigned to TEXTURE class.

Case 4: Tex$E > \mu_3$. If (15) is met for $\alpha = \kappa\alpha_1$ and $\beta = \kappa\beta_1$ (where $\kappa < 1$), the block is assigned to EDGE class; otherwise it will be assigned to TEXTURE class.

In this work, $\mu_1 = 125$, $\mu_2 = 290$, $\mu_3 = 900$, $\alpha_1 = 7$, $\beta_1 = 5$, $\kappa = 0.1$, and $\upsilon = 16$.

The extent of inter-band masking is measured as

$$\xi(n_1, n_2) = \begin{cases} 1 + [(\mathrm{Tex}E(n_1, n_2) \\ \quad -\mu_2)/(2\mu_3 - \mu_2)]\delta_1 & \text{for TEXTURE block} \\ \delta_1 & \text{for EDGE block} \\ & \text{and } L + M > 400, \\ \delta_2 & \text{for EDGE block} \\ & \text{and } L + M \leqslant 400, \\ 1 & \text{for PLAIN block} \end{cases} \tag{16}$$

where $\delta_1 = 1.25$ and $\delta_2 = 1.125$.

With due consideration of intra-band masking effect [11,14,33], we obtain the masking elevation factor:

$$a_e(n_1, n_2, i, j) = \begin{cases} \xi(n_1, n_2) & \text{for } (i,j) \in LF \cup MF \\ & \quad \text{in Edge block,} \\ \xi(n_1, n_2) & \text{otherwise,} \\ \times \max\left\{1, \left(\dfrac{C(n_1, n_2, i, j)}{t_b(n_1, n_2, i, j)}\right)^{\varepsilon}\right\} & \end{cases} \tag{17}$$

where $C(n_1, n_2, i, j)$ is the corresponding DCT coefficient, and $\varepsilon = 0.36$. Since the HVS is sensitive for the change on edge, the LF and MF portions for an EDGE block are excluded from the intra-band masking evaluation to avoid over-estimation of JND.

3. Use of JND in distortion evaluation and image compression

This section discusses how JND is adopted in perceptual visual distortion evaluation and DCT-based image encoding. The distortion metric and the image encoder presented will also be used in the experiments in Section 4 to demonstrate the performance of our JND model.

3.1. Perceptual distortion measure

Traditional error measures for images, such as mean square error (MSE) and peak signal-to-noise ratio (PSNR), do not provide a satisfactory reflection of the HVS' perception on image quality in many situations [6,7,27]. For better perceptual error evaluation, a JND-based subband distortion estimator can be defined as [33]:

$$d(n_1, n_2, i, j) = e(n_1, n_2, i, j)/t_{\mathrm{JND}}(n_1, n_2, i, j), \tag{18}$$

where JND threshold $t_{\mathrm{JND}}(n_1, n_2, i, j)$ can be derived from our JND model or other models (e.g., DCTune [33]); $e(n_1, n_2, i, j)$ is the subband error (due to compression or other processing). Firstly, Minkowski distance, which is a standard method in vision models to combine the probabilities for individual errors that are to be seen [24], is calculated in the same subband (i,j) over all blocks, as the *spatial error pooling*:

$$p(i,j) = \left(\sum_{n_1,n_2} |d(n_1,n_2,i,j)|^{\beta_p} \right)^{1/\beta_p}, \tag{19a}$$

where $\beta_p = 4$. For block-based transform coded images, the *spatial error pooling* can take an alternative form:

$$p(i,j) = \begin{cases} \left(\sum_{n_1,n_2} |d(n_1,n_2,i,j)|^{\beta_l} \right)^{1/\beta_l} & \text{for } (i,j) = (0,0), \\ & (0,1),(1,0), \\ \left(\sum_{n_1,n_2} |d(n_1,n_2,i,j)|^{\beta_p} \right)^{1/\beta_p} & \text{otherwise,} \end{cases} \tag{19b}$$

where $\beta_l = 2.3$. β_l is chosen lower than β_p to make the resultant distortion metrics more sensitive to blockiness artifacts with smooth areas in the block-based transform coding schemes.

We then perform the *frequency error pooling* to result in a single-valued perceptual distortion measure:

$$P = \left(\sum_{i,j} p(i,j)^{\beta_p} \right)^{1/\beta_p}. \tag{20}$$

Better JND estimation facilitates the perceptual distortion measurement to be closer to the HVS' perception for an image.

3.2. Prediction correlations for distortion metrics

Pearson correlation and *Spearman rank-order correlation* can be adopted to demonstrate the ability of a metric to predict subjective test scores. In order to facilitate monotonicity of prediction and a common analysis space of comparison, P' is determined by fitting the metric's output P against the associated subjective test score z [31] as:

$$P' = a_0 + a_1(P) + a_2(P^2) + a_3(P^3), \tag{21}$$

where a_0, a_1, a_2 and a_3 are determined by fitting P–z for all images under test.

Pearson correlation r_p is defined as follows to measure the prediction accuracy:

$$r_p = \frac{\sum_k (P'_k - \bar{P}')(z_k - \bar{z})}{\sqrt{\sum_k (P'_k - \bar{P}')^2}\sqrt{\sum_k (z_k - \bar{z})^2}}, \tag{22}$$

where $\bar{P}'$ and $\bar{z}$ are the means of P' and z, and k is the index for the image under test.

Spearman rank-order correlation r_s is defined as follows to measure the prediction consistency:

$$r_s = \frac{\sum_k (\chi_k - \bar{\chi})(\gamma_k - \bar{\gamma})}{\sqrt{\sum_k (\chi_k - \bar{\chi})^2}\sqrt{\sum_k (\gamma_k - \bar{\gamma})^2}}, \tag{23}$$

where χ_k is the rank of P'_k and γ_k is the rank of z_k in the ordered data series, and $\bar{\chi}$ and $\bar{\gamma}$ are the respective midranks. The higher the values of r_p and r_s the stronger association exists between metric output and the subjective test score. In the ideal match between the metric's output and subjective test scores, $r_p = 1$ and $r_s = 1$ should be obtained.

3.3. JND-aided image compression

In a subband image coder [1,3,9–11,22,25,29,30,33,34], data in each subband are quantized before source coding. Since quantization is the major source of signal distortion, a quantizer designed according to the JND map can minimize the visual distortion for a given compression rate. With JND, the quantization step can be straightforward set as

$$q(n_1,n_2,i,j) = 2t_{\text{JND}}(n_1,n_2,i,j) \tag{24}$$

to ensure that the maximum quantization error $q(n_1,n_2,i,j)/2$ would not exceed the corresponding JND threshold. The problem here is that $q(n_1,n_2,i,j)$ varies over the whole image, and thus results in large sided information that is to be transmitted to the decoder. One way to solve this problem is to derive a global quantization matrix QM [33] according to the JND map: QM $= \{\hat{q}(i,j)\}$ for all DCT blocks in an image. The difference between QM and the JPEG recommended luminance quantization array [8] is that

QM is image dependent and each quantizer is fine adjusted so that the compressed image can achieve the lowest perceptual distortion with the allowed bits.

We aim at the minimization of the perceptual subband distortion $P(i,j)$ defined in Eq. (19). $P(i,j)$ increases with $\hat{q}(i,j)$. Therefore, for each of the 64 DCT subbands, we set a target distortion $D_T(i,j)$, and then find a maximum $\hat{q}(i,j)$ to make sure that $P(i,j) \leqslant D_T(i,j)$, where $D_T(i,j)$ is inversely related with the number of allowed bits, Γ, for the image. The pseudocode for quantizer determination is shown in Fig. 5.

The final $\hat{q}(i,j)$ is the optimized quantizer for subband (i,j). To be consistent with JPEG, $\hat{q}(i,j)$ takes an integer within 1–255.

```
Decide an initial {D_T(i,j)}
Assume an initial QM (can be set to 1 for all elements)
Do{     increase{D_T(i,j)}
        for each subband (i,j)
                {       Do{calculate P(i,j) according to Equation (21-22)
                             increase q̂(i,j)
                          } while (P(i,j)≤ DT(i,j))
                }
    } while (the number of coded bits > Γ)
```

Fig. 5. Pseudocode for quantizer determination.

4. Performance of our JND model

4.1. Experiments on the base threshold and elevation factor

In order to compare the performance of our JND model and the DCTune model, two groups of experiments were conducted. The first group of experiments is designed to validate the base threshold in our model. In this group of experiments, Gaussian noise was added to dark, bright and midrange gray level areas (8×8 blocks) of *Actor* and *Lena* images with 256×256 8-bit format. Table 1 lists the 12 experimental conditions.

Subjective viewing was conducted to assess perceptual quality of the distorted images. Ten observers, 3 females and 7 males aged 20–27, were involved in the experiments. Their eyesight is either normal or has been corrected to normal with spectacles. Five of them have an experience on image processing and are familiar with the test images. The observers were asked to view the images on the same monitor (NEC MultiSync FE770) in a room illuminated with fluorescent ceiling lights with a viewing distance $\lambda = 50\,\mathrm{cm}$. At each time two images were shown on the screen: the original image on the left as the reference, and the noise-injected image on the right. The observers gave the noise-injected image a quality

Table 1
Generation of Gaussian-noise-injected images

Case #	Image	Range of average gray level for noise injection	Variance of Gaussian noise	Number of blocks injected with noise	Actual average gray level for affected blocks	MSE	Average subjective test z-scores
A1	Actor	0–40 (dark regions)	130.1	220	17.5	20.56	−0.7224
A2	Actor	0–90 (dark regions)	78.0	476	42.2	32.12	−0.2544
L1	Lena	0–75 (dark regions)	32.5	173	58.8	5.57	−1.1277
L2	Lena	0–75 (dark regions)	65.0	173	58.8	11.20	−0.3017
A3	Actor	85–150 (mid-range)	97.5	449	118.2	42.86	0.8822
A4	Actor	110–160 (mid-range)	65.0	317	135.9	20.09	0.6256
L3	Lena	110–160 (mid-range)	32.5	445	138.9	14.13	0.6487
L4	Lena	110–150 (mid-range)	65.0	326	132.0	20.71	1.5538
A5	Actor	160–255 (bright regions)	97.5	55	176.4	5.20	−0.6333
A6	Actor	170–255 (bright regions)	65.0	29	188.8	1.78	−0.5907
L5	Lena	170–255 (bright regions)	32.5	124	190.2	3.90	−0.3119
L6	Lena	170–255 (bright regions)	65.0	124	190.2	8.06	0.2318

Table 2
Description of subjective scores for the images under evaluation

Subjective score	Description
0	SAME quality as the original image
1	VERY GOOD quality although some difference may be spotted
2	GOOD quality
3	ACCEPTABLE quality
4	POOR quality
5	UNBEARABLE quality

score 0–5, as listed in Table 2, after viewing the display for at least 4 s. The order of the noise-injected images to be displayed for each observer was randomized so that the possible bias due to viewing experience was minimized. Prior to averaging the scores over all the 10 observers, we used *z-score* transform [5] to minimize the variation between the individual quality scores, since an observer may not use the full range of the numerical scale in classifying images. A *z*-score with an observer can be calculated as

$$z^t = \frac{s^t - \bar{s}}{\sigma}, \tag{25}$$

where s^t is the subjective score of tth test image, $\bar{s}$ is the mean score for all the test images for the observer, and σ is the standard deviation.

Perceptual distortion P for the noise-injected images has been measured via Eq. (20), using JND calculated by our model and the DCTune model.[3] Fig. 6(a) shows the scatter plot for P with the said models against the subjective scores. For the DCTune model, a poor estimate of the distortion measure P results with two cases (A1 and A2) of noise injection in very low gray-level (dark) regions. This is because the JND value in the DCTune model is very low in dark regions so that the resultant distortion measure is exaggerated through Eq. (18). The other two similar cases (L1 and L2, whose actual average gray levels of the noise-injected blocks are higher than A1 and A2) also lead to deviated data for the DCTune model. Fig. 6(b) gives a *zoom-in* version of the scatter plot with A1 and A2 excluded. From Fig. 6, it is obvious that our JND model is better in matching the subjective perception, especially in dark regions (this is exactly what we have devised for the base threshold, as Fig. 3).

In the second group of experiments, the elevation factor for contrast masking in our model is to be validated. With this aim, noise is added to DCT coefficients throughout an image according to the obtained JND:

$$C'(n_1, n_2, i, j) = C(n_1, n_2, i, j) + \tau S^{\text{random}}_{n_1, n_2, i, j} \times t_{\text{JND}}(n_1, n_2, i, j), \tag{26}$$

where $S^{\text{random}}_{n_1,\ n_2,\ i,j}$ takes $+1$ or -1 randomly regarding n_1, n_2, i and j, to avoid introduction of fixed pattern of changes, and τ is an adjustable parameter to ensure the same amount of error energy (therefore same MSE) among different JND estimators. This provides a way to evaluate how useful is the JND estimator. Under the same error energy, better visual quality of the image resulting from $C'(n_1, n_2, i, j)$ indicates a better JND estimator (especially in edge areas).

Noise-contaminated *Actor* and *Lena* images were created by Eq. (26) with the JND values derived from our model and the DCTune model, respectively. As listed in Table 3, Cases A7 and L7 were obtained via the DCTune model, while Cases A8 and L8 were obtained via our model with the same error energy (MSE). As shown in Fig. 7, A7 and L7 have more obvious distortions[4] than A8 and L8, especially around the boundaries of objects (e.g., both hands and the left sleeve in *Actor*, the hat rim and the right shoulder in *Lena*). Our scheme models more accurate JND around edge regions because of the block classification. This has been further demonstrated by the subjective viewing scores yielded through the same subjective test procedures described above. Fig. 8 shows that our scheme results in a better match with the subjective scores in perceptual distortion measurement than the DCTune model, for the cases illustrated in Table 3.

The first part of Table 4 compares *Pearson correlation* r_{p} and *Spearman correlation* r_{s} for the

[3]DCTune software has been downloaded via http://vision.arc.nasa.gov/dctune/#demo.

[4]Being even more obvious if they are compared on a monitor.

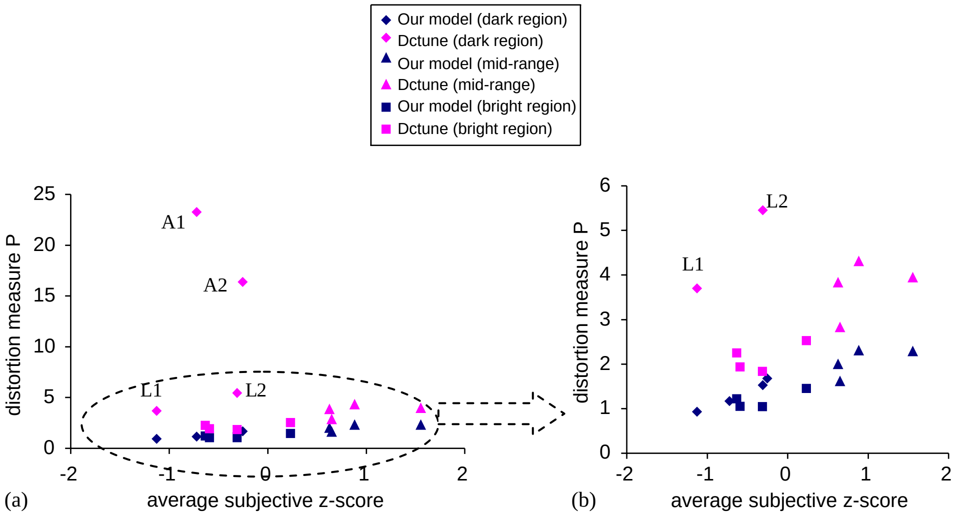

Fig. 6. Scatter plot for the two distortion measures against the average subjective scores (for the cases in Table 1).

Table 3
Generation of noise-contaminated images via Eq. (26)

Case #	Image	JND model used in Eq. (26)	MSE	Average subjective test z-score
A7	Actor	Watson's	135.86	0.8506
A8	Actor	Ours	135.86	0.0996
A9	Actor	Watson's	243.83	1.5973
L7	Lena	Watson's	88.53	0.9160
L8	Lena	Ours	88.53	−0.3838
L9	Lena	Watson's	151.04	1.4524

perceptual distortion measures facilitated by our JND scheme and the original DCTune model. For the subset of images with noise-injection in regions of bright and mid-range gray levels, the metric based on DCTune correlates reasonably well with the subjective scores. However, it does not correlate satisfactorily when the full set of 18 images (A1–A9 and L1–L9) is evaluated. As expected, the metric based upon our model performs well for both the subset and the full set of images, because of the improvement for luminance adaptation at dark regions and contrast masking around edges.

4.2. Experiments on perceptual image coding

Four 256×256 gray images (*Cameraman*, *Mandrill*, *Lena* and *Actor*) were compressed by using standard JPEG method and the JND-aided compression method presented in Section 3.3. For the latter, both our JND model and the DCTune model were tested, respectively. The three schemes compress each image at the same compression ratio (i.e. *bpp*—bits per pixel), as in Table 5.

The standard JPEG scheme has significant blocking artifacts in smooth areas. The JND-aided compression method with the DCTune model performs better in smooth areas, but causes some detail loss in the images. Compression with our JND model achieves better visual quality by preserving more details without causing severe blocking artifacts at the same *bpp*.

To prove this, we performed subjective viewing tests for the 12 decompressed images, with participation of 12 observers (5 female and 7 male). Their

Fig. 7. Noise-contaminated images: (a) Case A7 (noise injection according to Watson's model); (b) Case A8 (noise injection according to our model); (c) Case L7 (noise injection according to Watson's model); (d): Case L8 (noise injection according to our model).

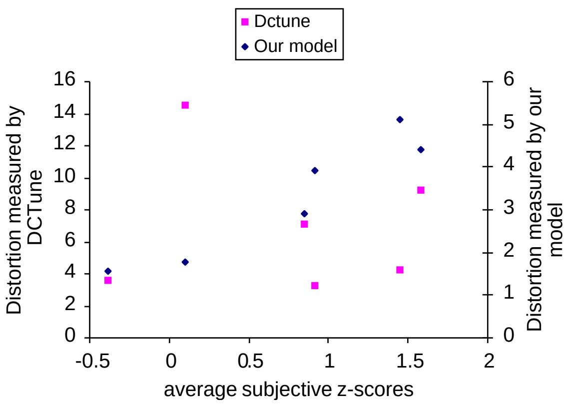

Fig. 8. Scatter plot for the two distortion measures against the average subjective scores (for the cases in Table 3).

eyesight is either normal or has been corrected to be normal. Six of the observers have experience on image processing and are familiar with the images. Each time two decompressed images from the same original were shown on the screen, and each observer was requested to indicate which image has better visual quality (or if they have the same quality). The order of displaying the decompressed image pair was randomized for each observer. The minimum viewing time for each display is 4 s. This procedure is similar to that in [16], and actually is a simplified version of the *difference-in-quality* method [5], which is suitable for comparison of multiple coders (with different coding artifacts) and different scenes.

Table 4
Correlations for the two distortion metrics

	Experiments	Metric	r_p	r_s
Experiments for base threshold and elevation factor	Sub-set of images (A3–A6; L3–L6) with only noise-injection in regions of bright and mid-range gray-levels	DCTune	0.9034	0.7619
		Ours	0.9252	0.8571
	Full set of images (A1–A9; L1–L9)	DCTune	0.6203	0.6202
		Ours	0.8393	0.7858
Experiments for coded images	Sub-set (9 compressed images) with exclusion of the *Actor* images	DCTune	0.8412	0.8333
		Ours	0.8735	0.8000
	Full set (12 compressed images)	DCTune	0.4255	0.3077
		Ours	0.8985	0.8392

Table 5
Compression ratio for different images

Image	*bpp*
Cameraman	0.27
Mandrill	0.30
Lena	0.26
Actor	0.33

In the procedure described above, each compressed image was compared for 24 times and the total times for which it was not chosen as the better image in the image pair serves as the *relative distortion indicator* for the corresponding compression scheme. Fig. 9 shows the subjective test results and confirms that the compression using our JND model outperforms the other two schemes, because it results in lowest *relative distortion* in the subjective tests among the three coding schemes.

The second part of Table 4 compares *Pearson correlation* r_p and *Spearman correlation* r_s for the perceptual distortion measures facilitated by our JND model and the original DCTune model. As can be seen, the metric based on DCTune only performs reasonably well for the subset excluding the coded *Actor* images. This is because *Actor* contains more dark areas than the other three images and DCTune does not work properly in those areas. As expected, the metric based upon our model performs well for both the subset and the full set of images.

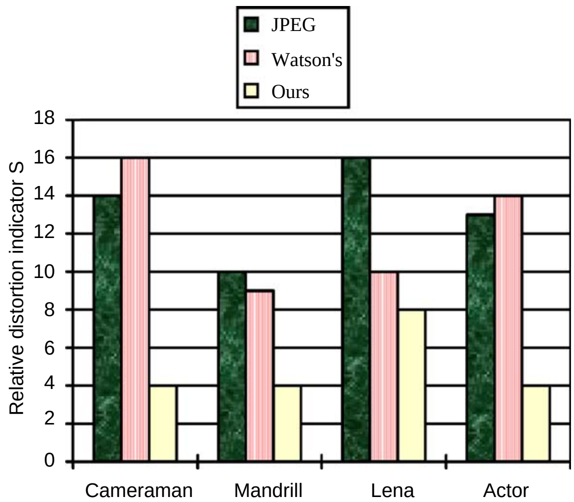

Fig. 9. Subjective test results for the three compression schemes.

5. Conclusion

An improved scheme for JND (just-noticeable distortion) estimation is proposed in this paper. It is proved to outperform the DCTune model for DCT subbands. The scheme accounts for the influence of spatial frequencies, luminance adaptation and contrast masking. The major contribution of this paper is a new formula for adjustment of luminance adaptation and incorporation of block classification in the model.

In digital images, the visibility threshold of the HVS (human visual system) exhibits an approximate parabola curve in the 0–255 range of gray levels. We formulate this phenomenon of local luminance adaptation to yield more accurate base threshold for JND. Contrast masking elevates the visibility threshold. However, edge regions need to be differentiated because the HVS' sensitivity is high in such regions. In our model, a lower elevation factor has been assigned to smooth and edge regions via block classification. This effectively avoids over-estimation of JND in the said regions.

Specially-designed experiments and the associated subjective tests demonstrate the improved performance of our JND model. The new base threshold proves better matching with the HVS perception for luminance adaptation (especially in dark regions) for digital images. The resultant elevation factor shows its improvement over edge regions. Our JND model has further shown to achieve favorable results in perceptual visual distortion gauge (closer matches with subjective viewing scores) and image compression (better quality for the same allowed bits).

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